

# Sri Lanka Institute of Information Technology



Lab Submission  
Lab sheet No 04

**IT24103586**

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**Probability and Statistics - IT2120**

## Exercise

- 1) Import the dataset ('Exercise.txt') into R and store it in a data frame called "branch data".

```
#1
# Set working directory
setwd("C:\\Users\\IT24103586\\Desktop\\IT24103586")
getwd()

# Import the dataset
branch_data <- read.table("Exercise.txt", header = TRUE, sep = ",")

> #1
> # Set working directory
> setwd("C:\\Users\\IT24103586\\Desktop\\IT24103586")
> getwd()
[1] "C:/Users/IT24103586/Desktop/IT24103586"
> # Import the dataset
> branch_data <- read.table("Exercise.txt", header = TRUE, sep = ",")
>
```

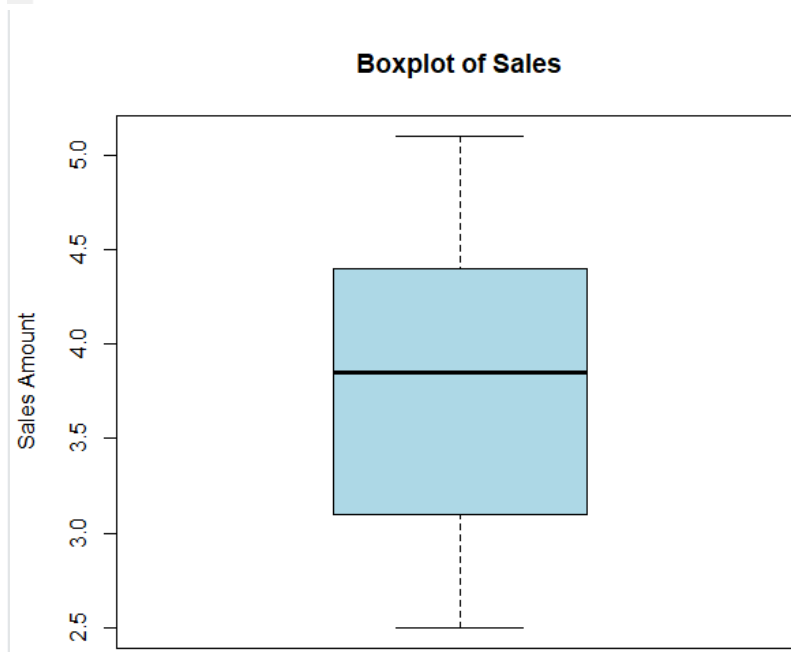
- 2) Identify the variable type and scale of measurement for each variable.

```
#2
# View the dataset
str(branch_data)

> #2
> # View the dataset
> str(branch_data)
'data.frame': 30 obs. of 4 variables:
 $ Branch      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Sales_X1    : num  3.4 4.1 2.8 5 3.7 4.5 3 4.9 3.2 2.5 ...
 $ Advertising_X2: int  120 150 90 200 110 175 95 185 105 80 ...
 $ Years_X3    : int  4 7 3 10 5 6 2 9 4 1 ...
```

- 3) Obtain boxplot for sales and interpret the shape of the sales distribution.

```
#3
boxplot(branch_data$Sales,
        main = "Boxplot of Sales",
        ylab = "Sales Amount",
        col = "lightblue",
        border = "black")
```



4) Calculate the five number summary and IQR for advertising variable.

```
summary(branch_data$Advertising_X2)
IQR(branch_data$Advertising_X2)
```

```
R 4.2.2 - C:/Users/IT24103586/Desktop/IT24103586/
> summary(branch_data$Advertising_X2)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 80.0   101.2   132.5   134.8   158.8   210.0
> IQR(branch_data$Advertising_X2)
[1] 57.5
>
```

5) Write an R function to find the outliers in a numeric vector and check for outliers in years variables.

```
#5
get.outliers <- function(z) {
  q1 <- quantile(z)[2]
  q3 <- quantile(z)[4]

  iqr <- q3 - q1

  ub <- q3 + 1.5 * iqr
  lb <- q1 - 1.5 * iqr

  print(paste("Upper Bound =", ub))
  print(paste("Lower Bound =", lb))

  outliers <- sort(z[z < lb | z > ub])

  if (length(outliers) > 0) {
    print(paste("Outliers:", paste(outliers, collapse = ",")))
  } else {
    print("No outliers detected.")
  }
}

get.outliers(branch_data$Years_X3)
```

```
R 4.2.2 - C:/Users/IT24103586/Desktop/IT24103586/
> #5
> get.outliers <- function(z) {
+   q1 <- quantile(z)[2]
+   q3 <- quantile(z)[4]
+
+   iqr <- q3 - q1
+
+   ub <- q3 + 1.5 * iqr
+   lb <- q1 - 1.5 * iqr
+
+   print(paste("Upper Bound =", ub))
+   print(paste("Lower Bound =", lb))
+
+   outliers <- sort(z[z < lb | z > ub])
+
+   if (length(outliers) > 0) {
+     print(paste("Outliers:", paste(outliers, collapse = ",")))
+   } else {
+     print("No outliers detected.")
+   }
+ }
> get.outliers(branch_data$Years_X3)
[1] "Upper Bound = 14.5"
[1] "Lower Bound = -3.5"
[1] "No outliers detected."
>
```